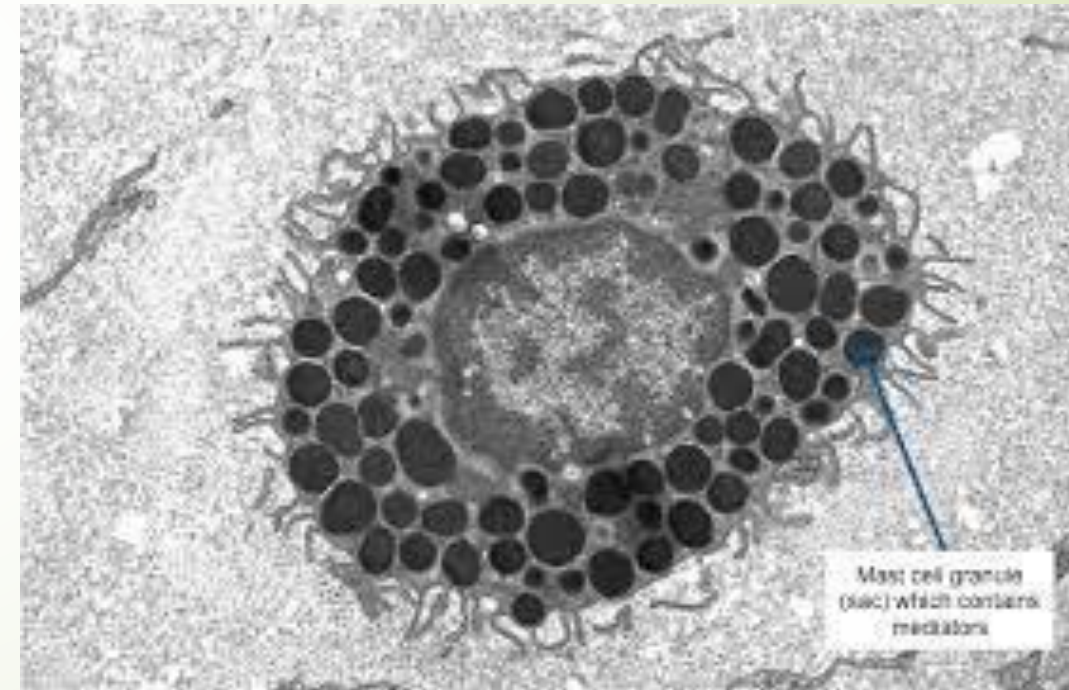
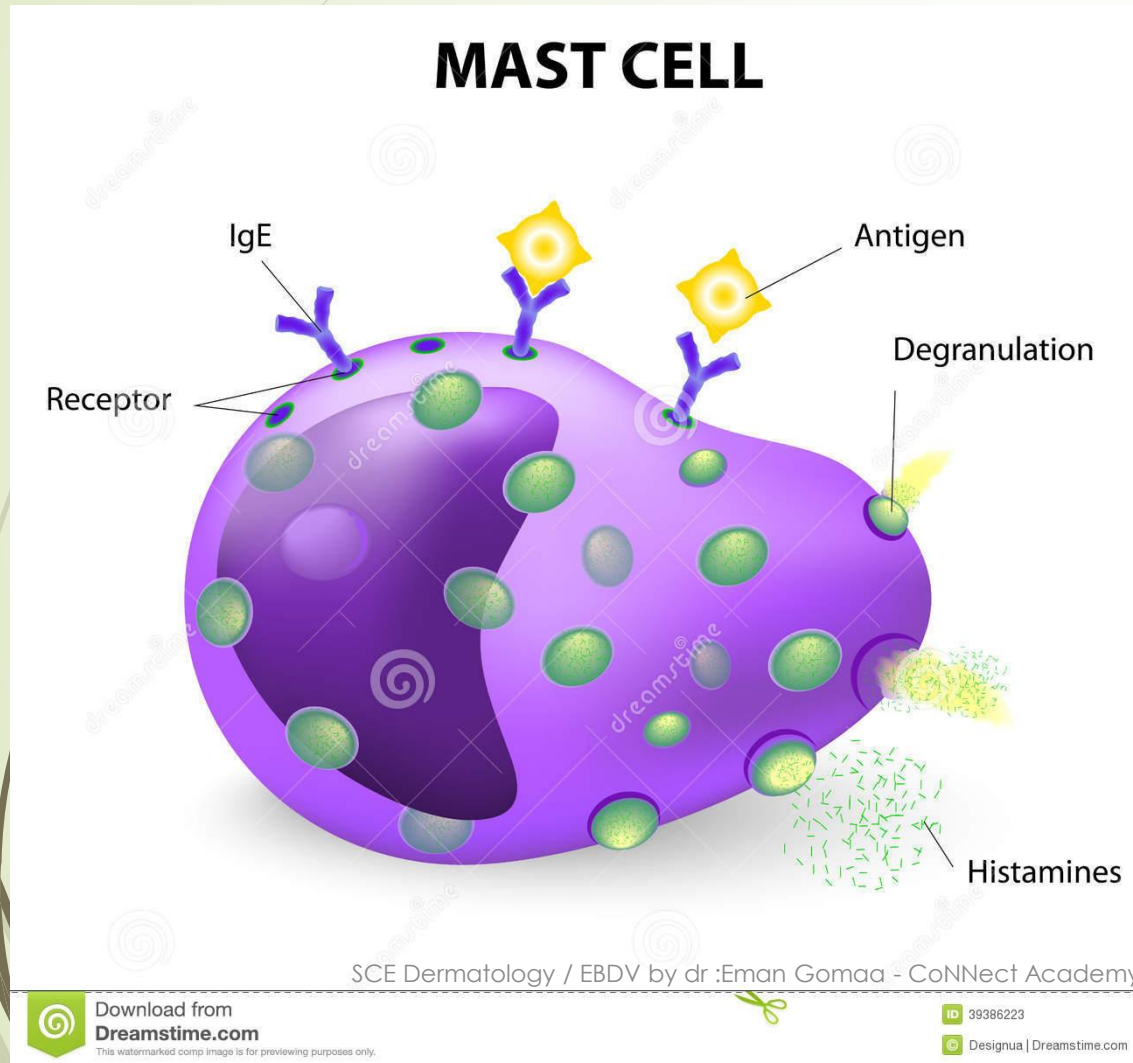


Mastocytosis , adnexal tumor , cyst

1

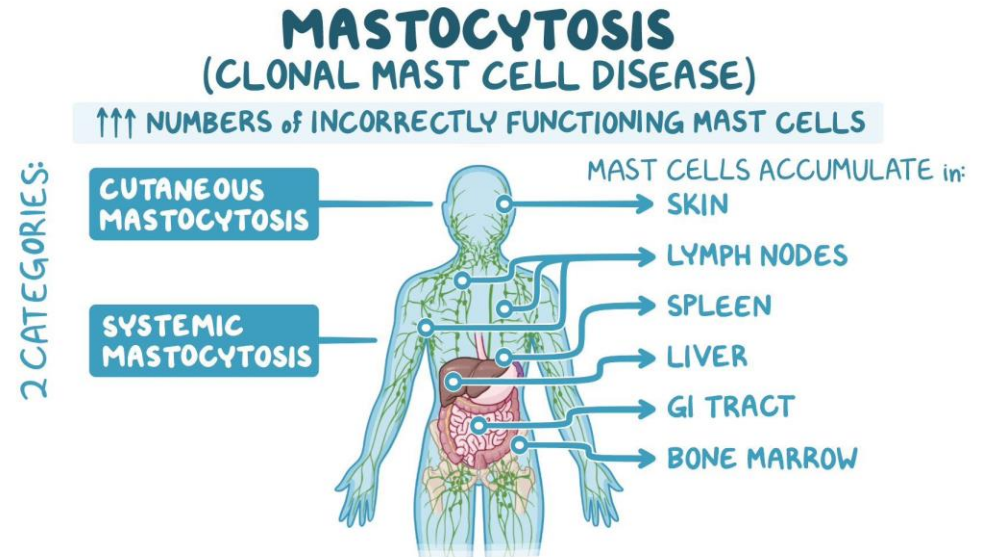
Dr eman gomaa
Consultant dermatologist

MASTOCYTOSIS



Mastocytosis

- Mastocytosis or mast cell disease represents a spectrum of clinical disorders with a common phenotype of tissue mast cell hyperplasia that can involve the skin, bone marrow, and other organs



4 Pathogenesis

Alterations in KIT structure and activity are central to the pathogenesis of mastocytosis.

Somatic activating mutations in KIT involving codon 816

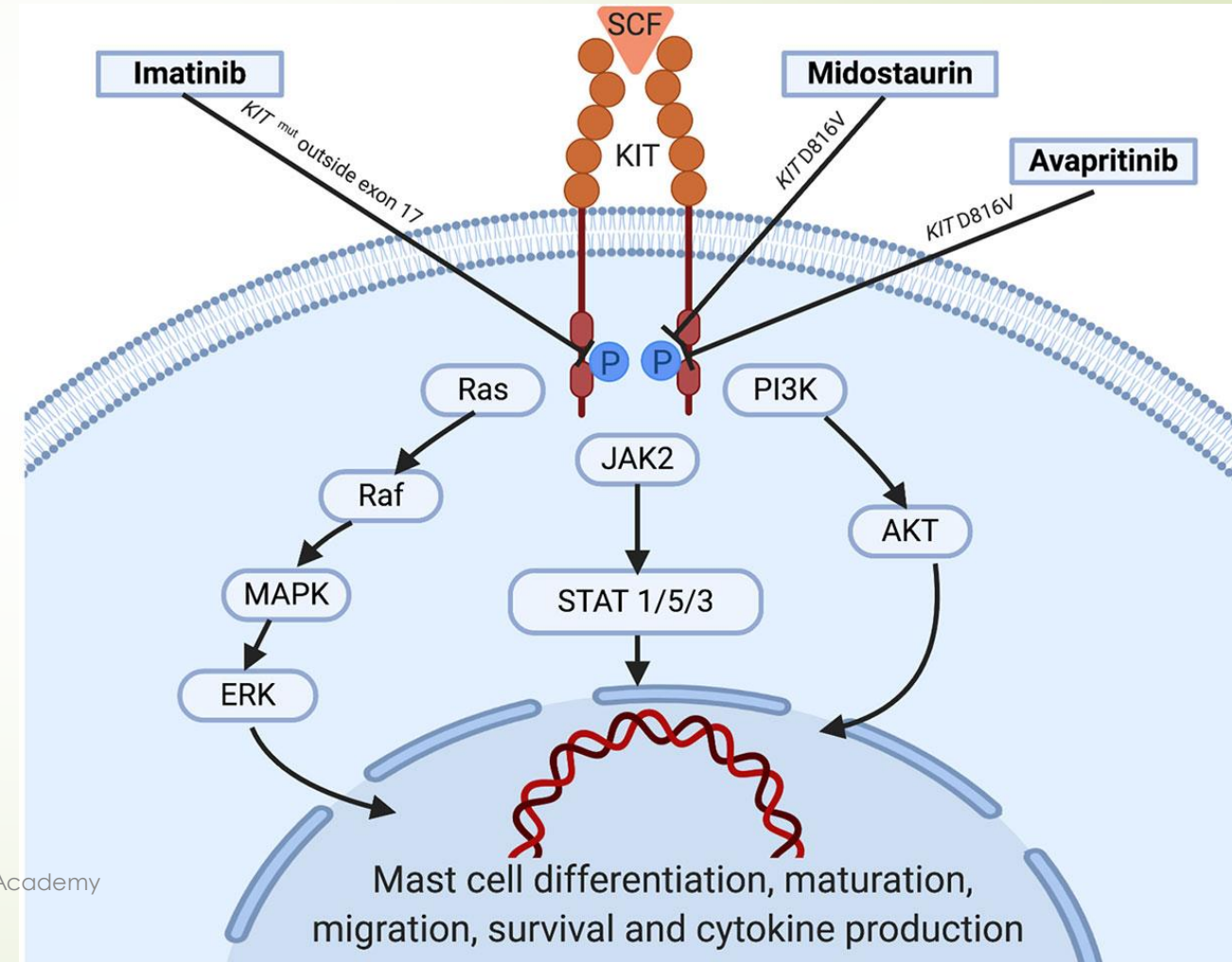
Activation of KIT by its ligand (stem cell factor [SCF]) induces mast cell growth and maturation

mastocytosis

- Mast cell →
- bone marrow → blood → CT → matur mast cell
- Cd117 – KIT TYROSINASE → it is activated by ligand stem cell
- autoactivating mutation in kit receptor → Out of control by ligand →
- →mastocytosis
-

Pathogenesis

- Somatic activating mutation in kit
- Ligand independent activation
- D816v , codin 618 , valin , aspartic acid



Clinical presentation of mastocytosis

Urticaria pigmentosa

mastocytoma

Diffuse
mastocytosis

Nodular
mastocytosis

Telangectasia
macularis
eruptiva
perestans

1-Urticaria pigmentosa

- Most common type 45%-75%
- two types:
 1. Childhood urticarial pigmentosa: favor trunk, spare face, palm, sole
 - * has 2 major variants : small (monomorphic brown macules or papules).
large (polymorphic yellow to brown plaques and macules).
 - * resolve spontaneously.
 2. Adult onset up: favor trunk and proximal extremities.
 - * Small monomorphic brown macules, papules
 - * associated with systemic mastocytosis



2-Mastocytoma:

- ❑ mainly in infancy , rare in adult.
- ❑ Solitary or few lesions <3 ,favodistal extremities
- ❑ Tan to brown nodule or plaque , has leathry texture



3-Diffuse mastocytosis:

- ❑ Childhood onset :first few monthes of life ,generalized ,hyperpigmented thickend skin with (leathery texture) ,good prognosis
- ❑ Adult onset : sever infiltrative mastocytosis especaly flexural site.
facial involvement may have leonine appearance



4- telangiectasia macularis eruptive perstans (tmeper)

- ❑ rare form of adult mastocytosis.
- ❑ Patches of telangiectasia without hyper pigmentation.



Bullous mastocytosis:

❑ Blistering following urticaria is common in infant and young children



****5- nodular mastocytosis**



- Systemic manifestation

- ☐ brain: headach , cognitive deregulation , fatigue
- ☐ GIT : abdominal pain , nausea , vomiting , diarrhea ,weight loss
- ☐ Cvs : palpitation ,chest pain , dyspnea , dizziness .
- ☐ Bone: pain

- * EXACERBATING FACTORS OF MASTOCYTOSIs:

- ☐ Friction , exercise , heat ,cold
- ☐ Spicy food , alcohol
- ☐ Medication : aspirin , NSAID , cough suppressant , lidocain

- Diagnosis:

- Darier sign :

- ❑ firmly stroking or rubbing of skin lesions induce urticarial like wheals that may accompanied by systemic symptom.
- ❑ mainly in mastocytoma>child hood up> adult up >tmep



➤ Histopathology:

- ❑ Perivascular mast cell infiltrate + melanophages + eosinophils
- ❑ H&E :fried egg appearance
- ❑ Special stains :leder stain ,toludin blue
- ❑ Immunohistchemistry:CD117(KIT)



1. Urticaria
2. Bullous in children
3. Nodular scabies

Investigations

- Serum tryptase
- Increased histamine level.
- Increased PG.
- Imaging.

- Avoidance of triggering factors.
 - Symptomatic:-
 - Topical :
 - Steroids
 - Calcineurin inhibitor
 - Cromolyn
 - Phototherapy :- NB, UVB & PUVA

Systemic

1. Antihistaminic
2. H2 antagonist
3. Mast cell stabilizer e.g. Zaditen
4. Omalizumab
5. Imatinibe:- Against kit receptor
Used in melanoma

WHO CRITERIA FOR THE DIAGNOSIS OF SYSTEMIC MASTOCYTOSIS

Requires either the major criterion plus one minor criterion or three minor criteria

Major criterion

Multifocal, dense infiltrates of mast cells (aggregates of ≥ 15 mast cells) in bone marrow or extracutaneous tissues

Minor criteria

1. $>25\%$ of mast cells in bone marrow samples or extracutaneous tissues are spindle-shaped or otherwise atypical
2. Expression of CD25 and/or CD2 by extracutaneous mast cells (often determined by bone marrow flow cytometry)
3. Presence of activating KIT codon 816 mutation in blood, bone marrow, or extracutaneous tissues
4. Serum total tryptase level persistently >20 ng/ml (unless there is an associated clonal myeloid disorder, in which case this parameter is not valid)

THERAPEUTIC LADDER FOR MASTOCYTOSIS

21

Avoidance of potential mast cell stimuli

Friction, Exercise, Heat (e.g. hot bath) or cold (e.g. swimming)

Hot beverages, Spicy foods

Aspirin, Nonsteroidal anti-inflammatory drugs, Narcotics (e.g. morphine, codeine)

Anticholinergics (e.g. scopolamine)

Local therapy for symptom control

Potent and superpotent topical corticosteroids, including under occlusion

(3) Intralesional corticosteroids

Systemic therapy for symptom control

Oral antihistamines

Oral cromolyn sodium

Omalizumab Oral PUVA [psoralens plus ultraviolet A]

Narrowband ultraviolet

Oral corticosteroids

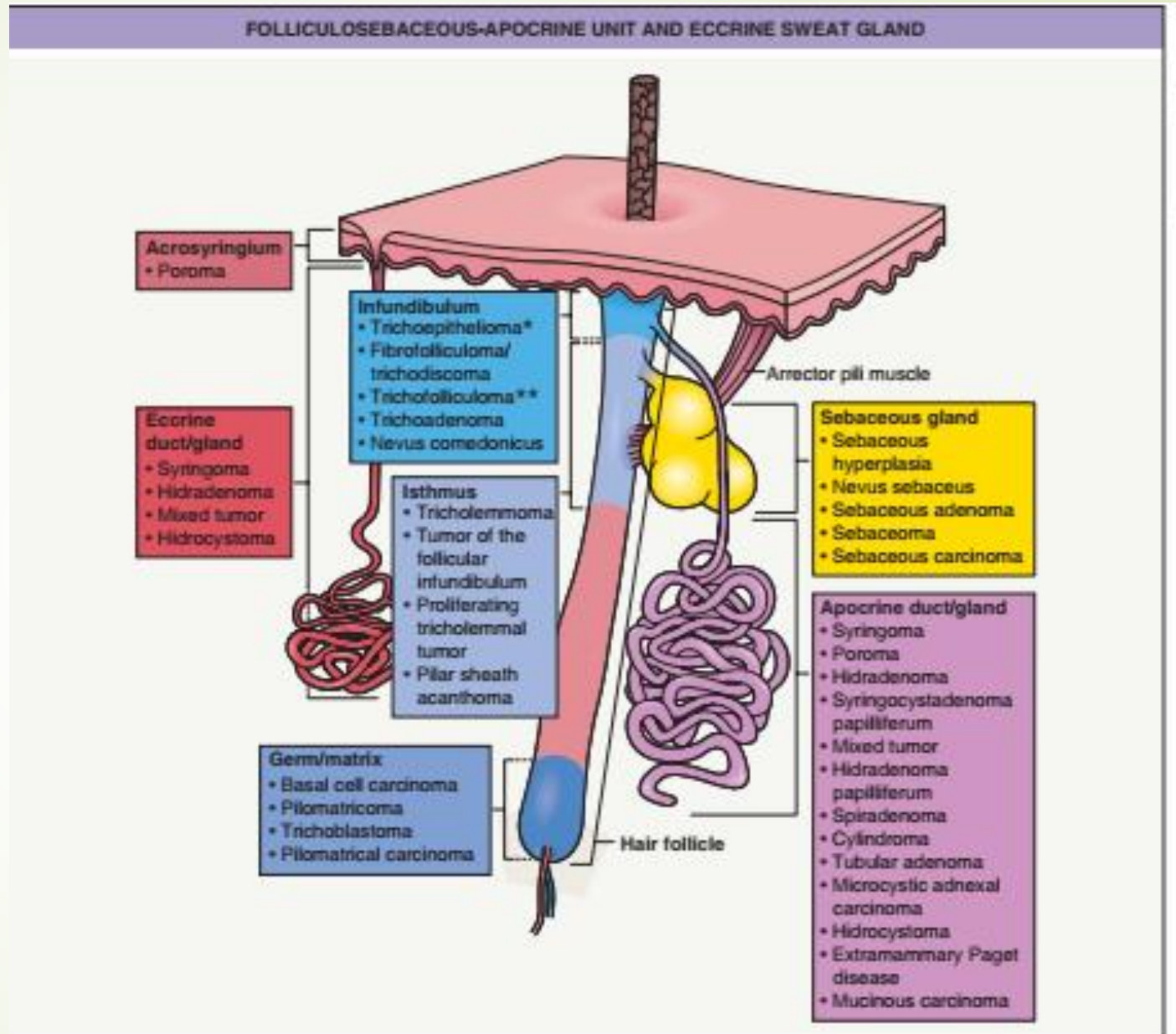
UVA1

Types of systemic mastocytosis

22

1. Indolent mastocytosis.
2. Aggressive systemic mastocytosis.
3. Mast cell leukemia.
4. Mast cell sarcoma.
5. Extra-cutaneous mastocytoma.

Adenxal tumor





- Infundibulum
- Face, scalp & upper neck.
- Papule, nodule → (Punctum) tuft of immature hair (villous follicle) → Immature.
 - **H.P.**
- Dilated infundibulum filled with orthokeratin → Inverted tree appearance.
 - **TTT**
- Surgical excision. (Not true tumor hamartoma)



Fig. 111.2
Trichofolliculoma. Wispy
vellus hairs emerge from
a skin-colored papule
with a dilated central

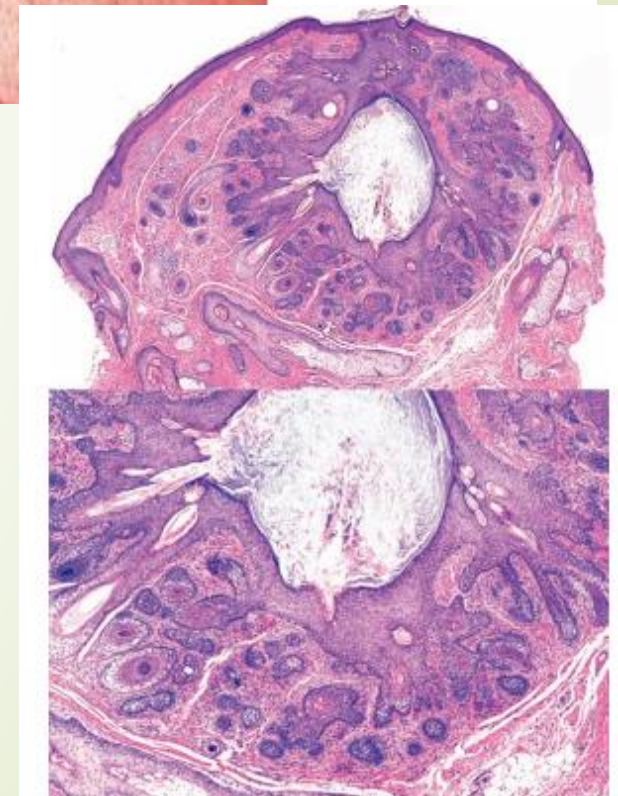


Fig. 111.3 Trichofolliculoma. There is a central patulous follicular infundibulum, from which fully formed or nearly fully formed follicles radiate. Courtesy, Luis Requena, MD.

EVALUATION OF PATIENTS WITH BIRT-HOGG-DUBÉ SYNDROME (BHDS)

Molecular genetic testing (*FLCN*) is recommended if patient is known to have BHDS or is suspected of having BHDS, including those who have one of the following:

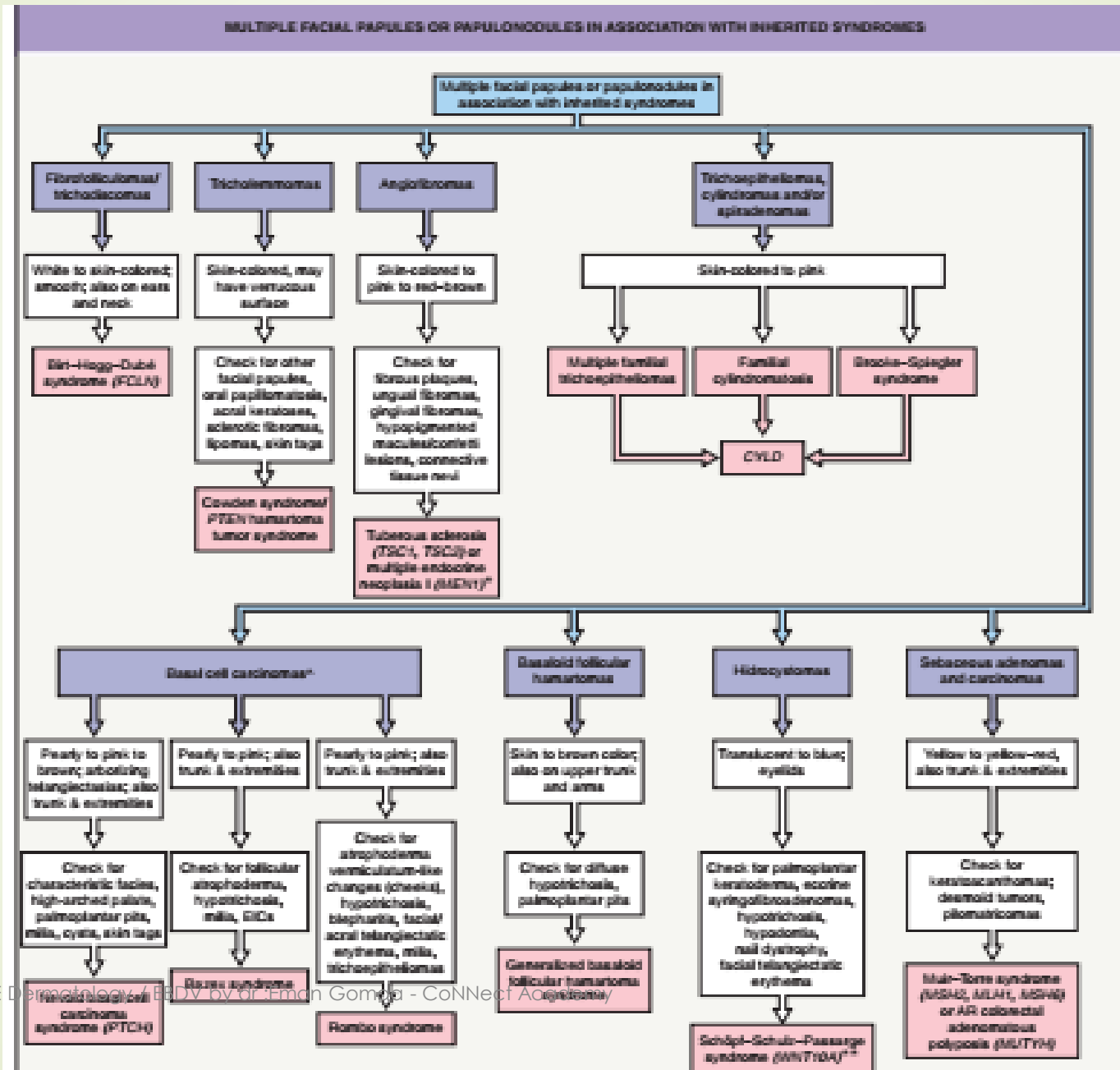
- ≥5 facial or truncal papules, at least one of which is confirmed histologically to be a fibrofolliculoma
- Facial papules with a histologic diagnosis of angiofibroma (excluding individuals with tuberous sclerosis or MEN1)
- Multiple and bilateral chromophobe, oncocytic, and/or hybrid renal tumors
- A single chromophobe, oncocytic, and/or hybrid renal tumor and a family history of renal cancer (any of these cell types)
- A family history of AD primary spontaneous pneumothorax but without a history of smoking or COPD

Imaging studies for pulmonary and renal involvement:

- Pulmonary: high-resolution CT scan of the chest to detect cysts
- Renal: renal MRI scan (preferred) or abdominal/pelvic CT scan to detect tumors

Suggested surveillance and recommendations:

- Cutaneous: full body skin examination
- Pulmonary: smoking cessation and judicious use of general anesthesia to reduce risk of pneumothorax
- Renal: annual renal MRI scan (preferred) or abdominal/pelvic CT scan, but if no family history of renal tumors plus 2–3 negative annual scans, screening can be done every 2 years
 - Tumors <3 cm in diameter monitored by periodic imaging
 - Tumors >3 cm in diameter treated with surgical excision



28 Trichoepithelioma

Nasolabial fold

Single or multiple.

H.P.:- LIKE BCC

Palisading

Cleft.

No follicular differentiation.

Basophilic baseloid cells

with tight stroma

With follicular differentiation.



Fig. 111.12 Multiple familial trichoepitheliomas. Numerous skin-colored papules and nodules on the midface. Patients have inactivating mutations in the tumor suppressor gene *CYLD*.

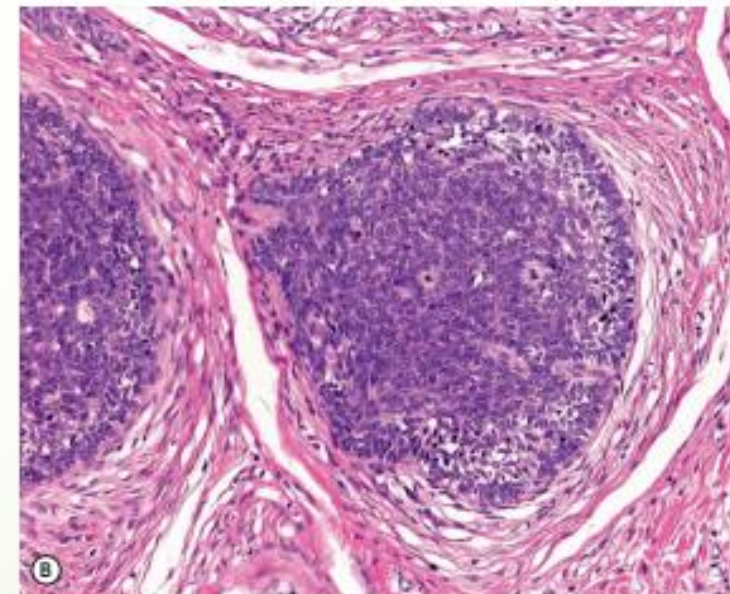
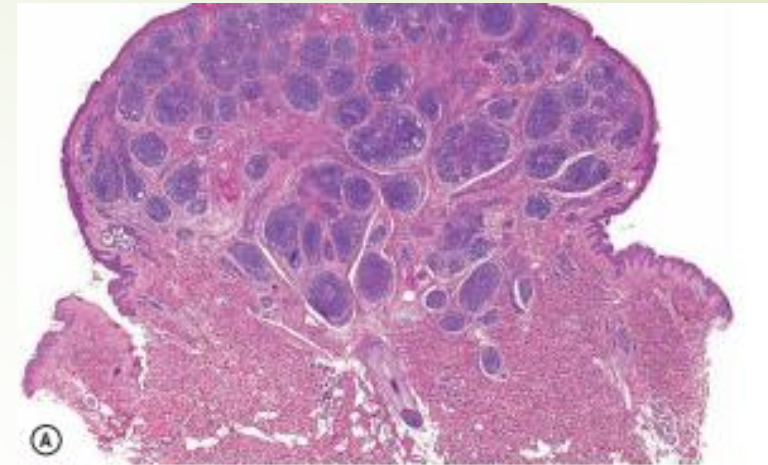


Fig. 111.13 Trichoepithelioma (trichoblastoma). **A** Dome-shaped papule composed primarily of clusters of follicular germinative cells. **B** A cluster of follicular germinative cells. Note that the clefts are between stromal elements, in contrast to the clefts between tumor and stroma that are characteristic of basal cell carcinoma.

29

Pilomatricoma

Childhood

Firm (Rock) hard nodule.

H.P.:-

Early → Mass of basaloid matrical cells (Ghost cells)

Late → Calcification & Ossification.



Fig. 111.15
Pilomatricoma. A firm
nodule on the cheek of a
child.

Sebaceous Tumors

Fordyce
spots
montgomery
tubercle

Sebaceous
hyperplasia

Nevus
sebaceous
Of tadassohn

31 Fordyce spots montgomery tubercle

Ectopic mature sebaceous gland.

Fordyce → Vermilian border of lips.

Montgomery → areola of breast

Association:- Ipsilateral facial paralysis.

Group of small but mature sebaceous lobules around small duct with direct opening to skin

NO TTT :- LASER, cautery and beeling



Sebaceous hyperplasia

32

Enlargement (without increase in number) of sebaceous gland.

Common in adult with immunodeficiency in transplant recipients.

Central up in the face.

C.Pic → multiple yellow papules with central depression



Nevus sebaceous Of jadassohn

33

Non neoplastic nevus or congenital malformation.
Include:- Seb-follicular, apocrine, epidermal
hyperplasia (Hamartoma)

C.Pic:- Scalp & Face

Rare → trunk → Linear.

4 Stages

Patch of alopecia at birth → Thick yellow Childhood
→ Verrucous at adolescence → 2ry adnexal tumor
benign > malignant (trichoblastoma).



Fig. 111.6 Nevus sebaceous. An oval yellow-pink verrucous plaque on the scalp. Note the alopecia.



Fig. 111.7 Syringocystadenoma papilliferum arising within a nevus sebaceous. Yellow-brown verrucous plaque in which an eroded, pink, lobulated papulonodule developed. Courtesy, Kalman Wasyk, MD.

H.P.:-

Hyperplasia

Epidermis

Sebaceous

Apocrine gland.

Organoid nevus

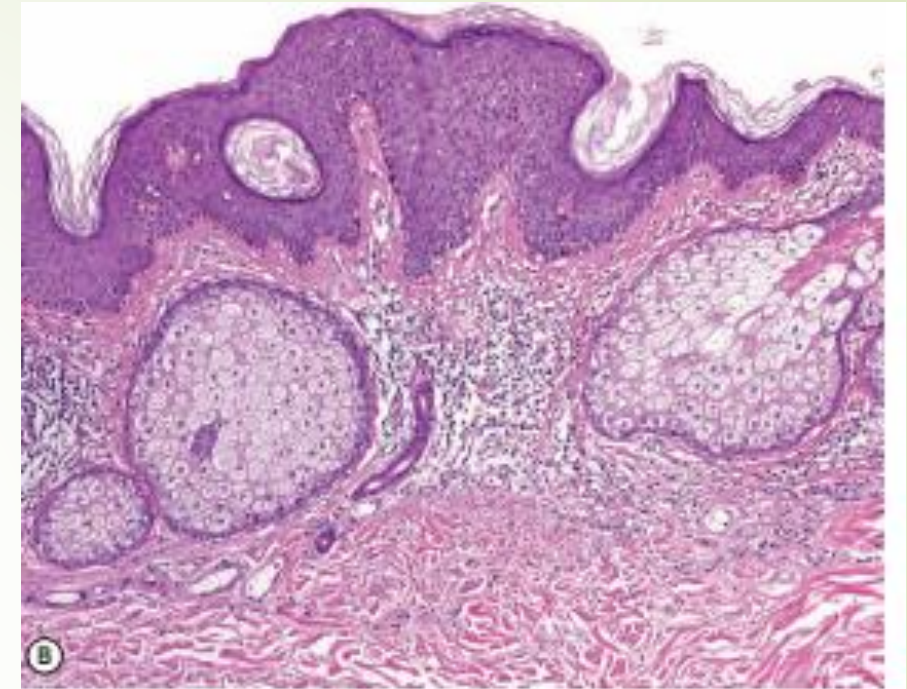
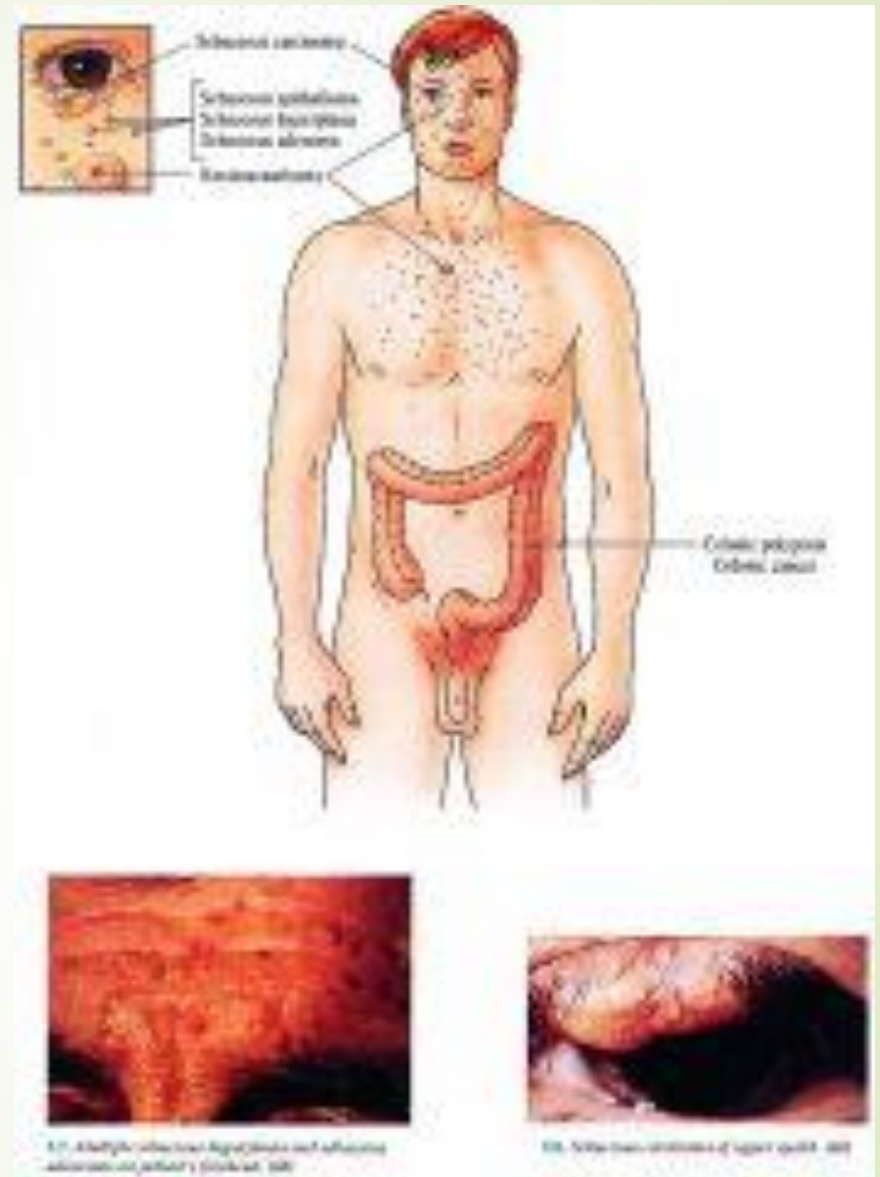


Fig. 111.8 Nevus sebaceus. A Hyperplasia of the epidermis in a papillated fashion, much like the pattern of an epidermal nevus. This is accompanied by adnexal malformation, with enlarged sebaceous lobules. Note the apocrine glands (arrow). B The coarsely vacuolated cytoplasm is diagnostic of sebocytes.

MUIR TORRE SYNDROM :
 Sebaceous adenoma
 KA
 Colorectal tumor
 Genitourinary



Syringoma

- Benign adnexal tumor with ductal differentiation
- of eccrine, apocrine lineage
- Skin colored flat topped papules
- Multiple or eruptive
- Periorbital
- Eruptive at trunk



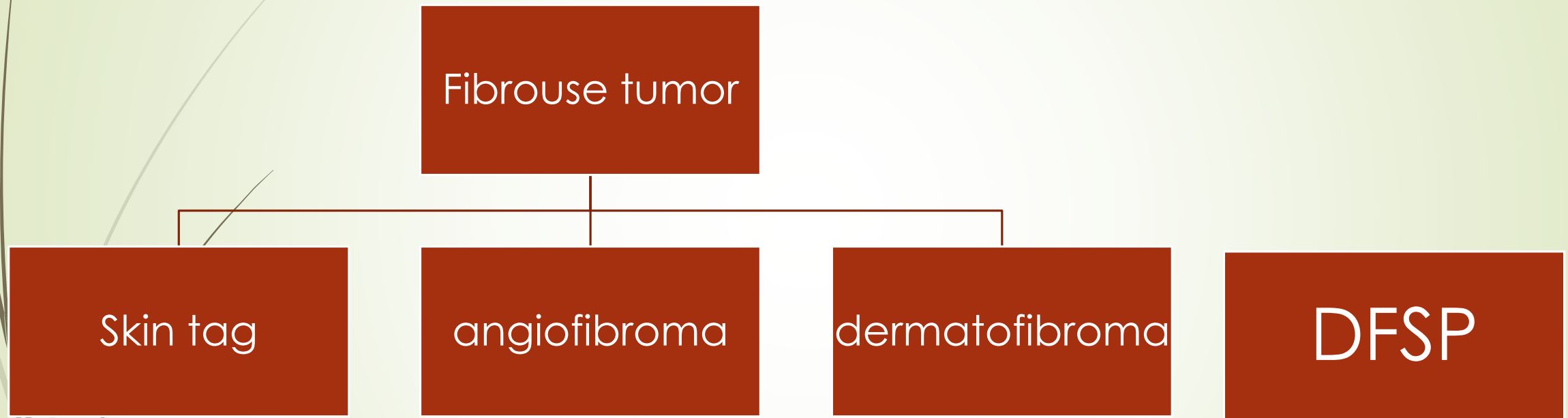
- H.P:
- Superficial multiple nest of cells with pale cytoplasm
- Central ductal differentiation
- Ecrin sweat duct at upper papillary dermis
- Tad pole appearance



- DD
 - Milia
 - Planwart
 - Xanthelasma



Figure 4: Milia's skin raised nodules over the chest and neck.



SKIN TAG

- Very common ,equal sex
- Risk factor :
 - Age , diabete ,obesity
 - Colonic polyp
 - Cvs
- Side of neck , axilla , groin
- Soft skin colored , pedunculated



Angiofibroma

➤ H.P term for group of lesions :

1. **Fibrous papule of face**
2. **Pearly penil papules**
3. **Multiple facial angiofibroma**
4. **Ungual fibroma**

koenen



Dermatofibroma sclerosing hemangioma

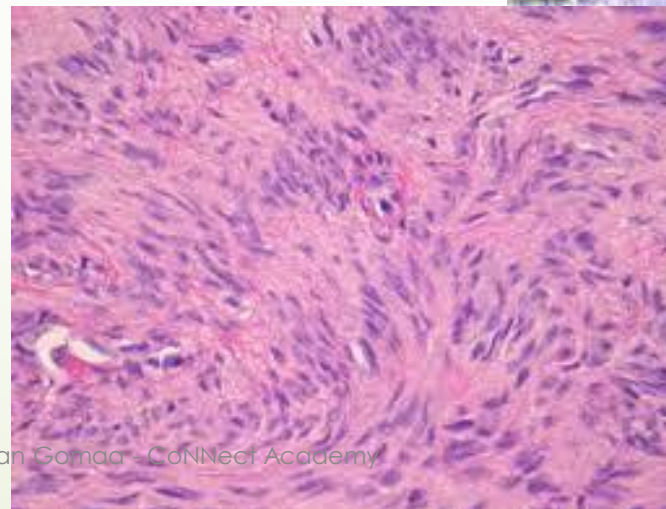
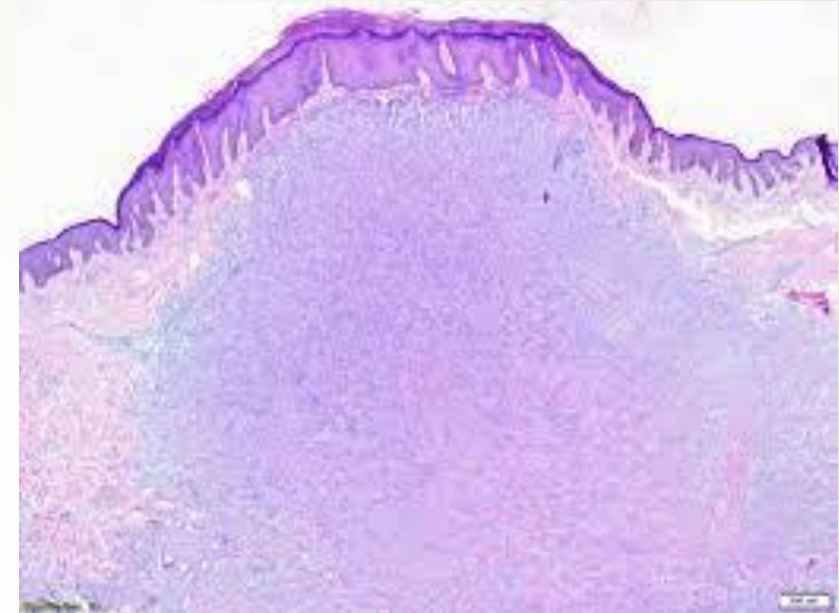
- Benign proliferation of
- fibroblast , histeocyt
- Fibrous histocytoma
- Trauma or insect bite
- Multiple DF :
 - FAMILIAL , LE , HIV



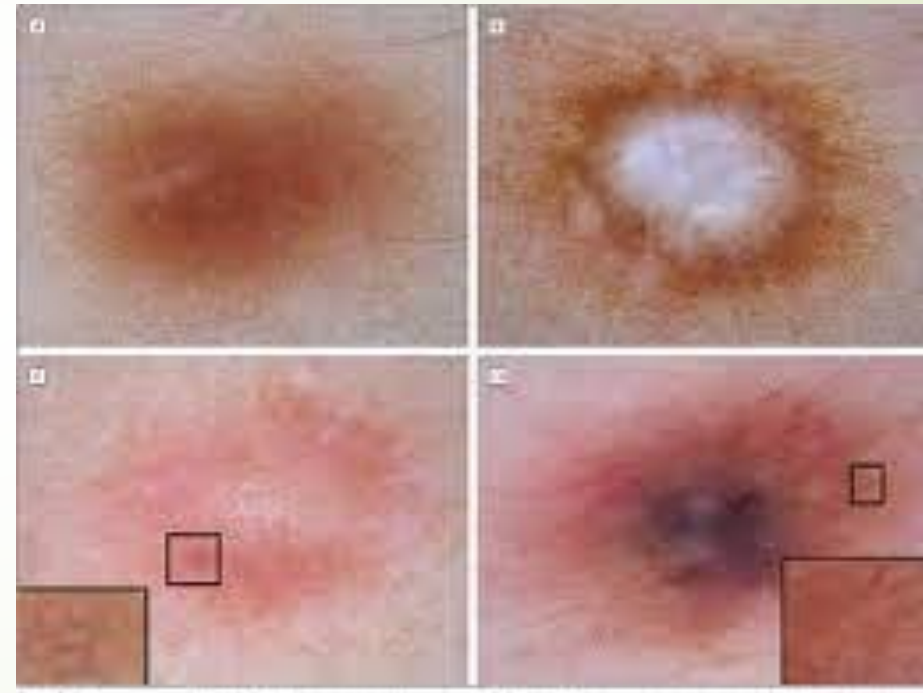
- Firm ,dome shaped , hyperpigmented papules
- Lower limb
- **Dimple sign**



- H.P
 - Fibroblast
 - Histiocyte
 - Hemosiderine
- **Factor x///a +ve**
- CD 34 -ve



- DERMOSCOPY :
 - Central white scar
- Tt:
 - No
 - Surgical excision



Dermatofibrosarcoma protuberance

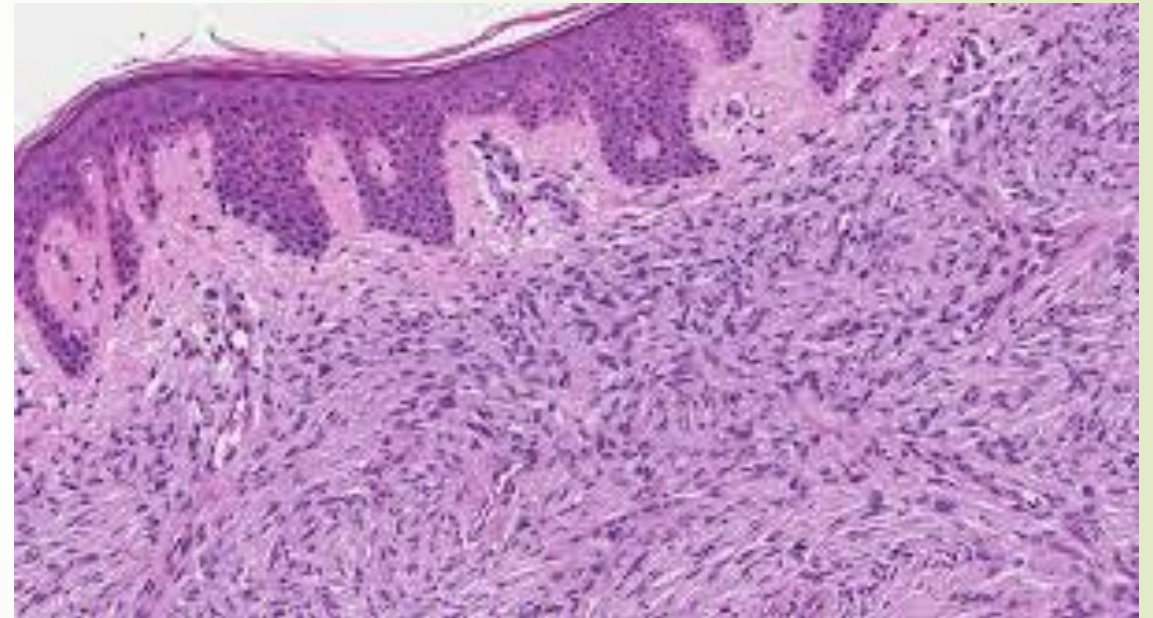
- Slowly growing locally aggressive sarcoma
- Rare but most common dermal sarcoma
- Intermediate malignancy
- Black women
- Shoulder , pelvic girdle



- Solitary multinodular mass
- Hard indurated base
- Better felt than seen
- Painfull ulceration , discharge
- Causes:
 - Trauma
 - Radiotherapy
- Local invasion , local recurrence



- Hp :
 - Spindle cell
 - Storiform mat like
 - Honey comb pattern
- **CD 34 +VE**
- FACTOR X/// α -ve



➤ TT:

➤ MOH MICRIGRAPHIC SURGRY

DD:

KELOID

MORPHEA

DF

Cutaneous epidermal cysts

Milia

1. Small, superficial
2. Common in all ages.
3. Infundibulum.

1ry

2ry

Burn, EB & PCT.

Bullous L.P.

Steroid atrophy

Follicular M.F.

Types



C.Pic:-

- White, yellow small dome shaped
- Subepidermal papule.
- Eye lid & cheeks.

Variants:- Eruptive &

- new born
- 40-50% in face
 - ✓ In gum → Pohnes nodules
 - ✓ hard → Epistien Pearls.
- Milia en Plaque.
- Milia syndrome.

BCC syndrome



Figure 9. Epstein Pearls

Epidermoid cyst

- Most common
- Originates from → infundibulum.
- Types → 1ry &
- 2ry.
 - Severe acne vulgaris.
 - Gardner syndrome.
 - Basal cell nevus syndrome.
 - B raf inhibitors



- Well demarcated yellow or skin colored dermal nodule with central punctum.
- Face & Upper trunk.
- Asymptomatic → rupture → inflammation.
 - Variants:-
 1. Verrucous cyst.
 2. Proliferating epidermoid cyst
 3. Pigmented follicular cyst.
 4. Vellus hair cyst.
- Multiple, trunk with → steatocystoma in pachynychia congenita



Tricholemmal cyst

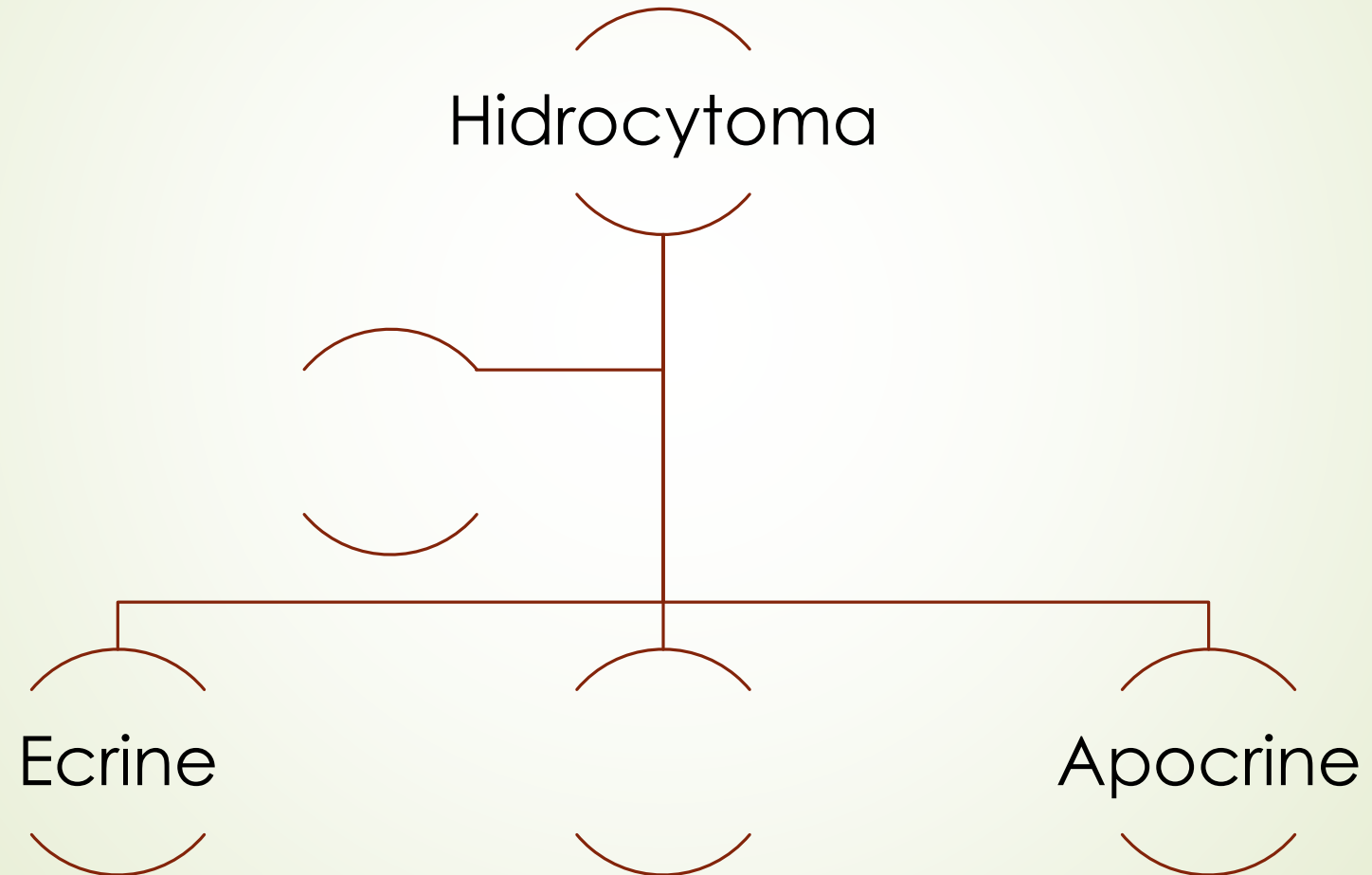
- Pilar cyst.
- Origin :- isthmus.
- Middle age women
- Common 5 Folds less epidermoid cyst
 - C.Pic:-
- Like epidermoid cyst (scalp)-multiple- smooth mobile firm nodule.
- H.P:- Wall SS epithelium (no granular layer)
- Cavity:- Compact keratin + Calcification.
- Treated by surgical excision.



Steatocystoma multiplex

- Dermal cysts.
- Wall:- sebaceous gland lobule , content sebum.
- Uncommon, at puberty both sex.
- Etiopathogenesis:- **AD, KRT 17 with pachynechia congenital**
- C.pic:- multiple smooth compressible nodules drain oil fluid.
- →trunk, proximal limb, presternal.
- Solitary, simplex and multiple.
- H.P.:- mid dermis wall :- SS epith no granular lined with thin irregular corrugated (eosinophilic cuticles)





Ecrine hidrocystoma

- Cystic dilatation of eccrine duct due to retention of fluid.
- Middle aged women increased with heat.
- Usually multiple, cystic & translucent.
- Cheeks & eyelids.
- H.p.-: unilocular cyst → 2 layers → cuboidal & flattened.
- Ttt:- simple excision
- Electrocautery.
- Multiple :- oxybutynin (uriban)
- Local atropine.



Apocrine

- Adenoma of apocrine gland.
- Adult & equal sex.
- Usually solitary.
- Translucent nodule.
- Outer canthus.
- Unilocular, multiple & local
- 2 layers → cuboidal high columnar → decapitation.
- TTT → same as eccrine hidrocystoma



A 64-year-old woman presented in the dermatology outpatient clinic with a 6-month history of solitary asymptomatic skin coloured nodule on her left cheek. It is gradually increasing in size, with no history of any pain or discharge. On cutaneous examination, there is a solitary nodule measuring 1.5 x 1.0 cm, firm to stony hard in consistency, non-tender easily mobile present over the left cheek. Tent sign is positive as the angulated shape can be appreciated by stretching the overlying skin. There is no cervical, occipital, or mandibular lymphadenopathy.

Histopathology revealed well-defined and encapsulated masses of basaloid cells with minimal cytoplasm that transition centrally into eosinophilic shadow or ghost cells containing no nucleus with prominent calcification.

What is the most likely diagnosis?

- ☐ Pilomatricoma
- ☐ Trichoblastoma
- ☐ Trichodiscoma
- ☐ Trichoepithelioma
- ☐ Trichofolliculoma

☒ Pilomatricoma ← Correct answer	82%
☐ Trichoblastoma	5%
☐ Trichodiscoma	2%
☐ Trichoepithelioma	10%
☐ Trichofolliculoma	1%

Explanation:

Pilomatricoma or calcifying epithelioma of Malherbe is a benign neoplasm composed of cells resembling the hair matrix, cortex, and inner root sheath. It can occur at any age with the majority occurring in the first or sixth decades. Clinically, this is presented as a solitary, deep dermal subcutaneous tumour. The characteristic histopathological findings include a lesion that is benign but malignant changes have been reported.

Trichoblastoma is a benign hair germ tumour composed of follicular germinative cells present in adults between the fourth and sixth decades. Clinically, this is presented as a solitary, well-defined, brown or black nodule on the head and neck. Histologically, trichoblastoma shows nests of well-circumscribed basophilic basaloid cells surrounded by sclerotic or hyalinised stroma.

Trichodiscoma is a benign tumour of the hair follicle mesenchyme. Clinically, it presents as multiple, discrete, 2-4 mm whitish flat-topped papules on the central face, upper trunk, and arms. They are rarely solitary and have been described in association with Birt-Hogg-Dube syndrome. Histologically, it shows fine fibrillary collagenous stroma having ectatic blood vessels with hair follicles at the margin.

Trichoepithelioma is a benign tumour showing follicular germinative differentiation. Clinically, it can be solitary or multiple. Solitary trichoepithelioma is a skin-coloured nodule usually on the face and resembles non-basal cell carcinoma. Multiple trichoepitheliomas are seen in association with Brooke-Spiegler syndrome. Histologically, there are lobules of monomorphic basaloid cells often with peripheral palisading surrounded by fibrous cellular stroma.

Trichofolliculoma is a benign neoplasm of pilosebaceous follicle. Clinically, it is recognised as a small dome-shaped nodule with a predilection for the face with two or three hairs protruding from the central pore. Histologically, it is composed of a large central follicle from which multiple small hair follicles emanate.

A 67-year-old woman presented to the skin cancer clinic, with a 15mm diameter asymptomatic plaque on her right lower leg. It had been slowly growing over the past year. On examination, she had Fitzpatrick type 1 skin. Dermoscopic findings are below.



Image: By Mohammad2018 - Own work, CC BY-SA 4.0

What is the most likely diagnosis?

- ☐ Actinic Keratosis
- ☐ Amelanotic melanoma
- ☐ Basal Cell Carcinoma
- ☐ Bowen's disease
- ☐ Squamous Cell Carcinoma

☐ Actinic Keratosis

8%

☐ Amelanotic melanoma

20%

☐ Basal Cell Carcinoma

9%

☒ Bowen's disease ← Correct answer

59%

☐ Squamous Cell Carcinoma

4%

Explanation:

The main dermoscopic findings here are erythema, keratosis and glomerular vessels. All of these features are seen in Bowen's disease. Glomerular vessels are coiled, convoluted capillaries, and are supposed to be reminiscent of the glomerulus of the kidneys. They correlate to dilated capillaries in the dermal papillae. Glomerular vessels are also commonly seen in squamous cell carcinoma. However, this lesion is described as a "plaque", whereas an SCC might be either more nodular or ulcerated. This lesion is also described as being asymptomatic and slowly growing over the course of a year, an SCC is more likely to be painful, and more likely to have grown quickly over the space of weeks-months. Squamous cell carcinomas may also have arborising vessels and hairpin vessels, which are not commonly found in Bowen's disease.

A 76-year-old Asian man seeks your advice regarding a recently developed asymptomatic red lesion on his forehead. He is a retired teacher and his leisure activities include gardening. His medical record shows that he underwent a renal transplant 5 years ago and is currently on immunosuppressive treatment. Physical examination revealed Fitzpatrick skin type 4, having a 1 x 1.5 cm red-coloured solitary non-tender firm nodule with a smooth shiny surface and overlying telangiectasias on the forehead. There is no evidence of any sun damage. There are no palpable lymph nodes in the head or neck area.

The patient is concerned as one of his elder sisters was recently diagnosed with basal cell carcinoma. Dermoscopic examination reveals the presence of telangiectasias with milky white and structureless areas. An excisional biopsy was taken and the histopathology result shows a dermal nodule consisting of small blue cells with numerous mitoses and a high chromogranin A, and synaptophysin.

Considering the neoplastic nature of the disease, further workup is planned to look for local and distant metastases.

What favourable prognostic factor is identified in this case?

☐ Fitzpatrick skin type 4

☐ Male gender

☐ Solitary lesion

☐ Tumour being localised to the face

☐ Tumour size less than 2 cm

☐ Fitzpatrick skin type 414%

☐ Male gender7%

☐ Solitary lesion10%

☐ Tumour being localised to the face3%

☐ Tumour size less than 2 cm ← Correct answer66%

Explanation:

Merkel cell carcinoma is a highly aggressive neuroendocrine skin cancer. Predisposing factors include exposure to UV radiation, old age, immunosuppression, and polyomavirus infection. It usually presents as a solitary red rapidly growing firm spherical tumour with a smooth shiny surface. It appears in the mid dermis as a nodular growth, solid sheet of cells, or shows a diffuse growth pattern.

Histopathologically, it is divided into 3 distinct types:

Types	Percentage	Features
Intermediate type	90%	Cells with medium-sized nuclei containing multiple small nucleoli and salt and pepper type chromatin
Small cell type	5%	Resemble metastasis of small cell carcinoma of lungs. Differentiated on the basis of IHC markers.
Trabecular type	5%	Shows epithelial growth pattern

Immunohistochemical markers are necessary for diagnosis and include CK 20, chromogranin A, synaptophysin and MCPyV T antigen. Some clinical prognostic factors with good outcomes include female gender, a size less than 2 cm, local disease, and a tumour localised to the upper limb while histological factors with poor outcomes include tumour size more than 5 cm, extension to subcutaneous tissues, diffuse growth pattern, high mitotic rate and absence of T cells infiltrate in the tumour.

A 66-year-old man complained of multiple scaly lesions on his scalp and face which had significantly increased over the past few years. He has worked as a farmer for more than 40 years. A physical examination of the patient revealed extensive, confluent hyperkeratotic plaques with surrounding erythema and background photodamage on the scalp, face, dorsal aspects of the hands and forearms, which were consistent with actinic keratoses. You consider field cancerization treatment which inhibits thymidylate synthase to reduce the formation of cutaneous squamous cell carcinoma.

Which of the following options best addresses the above mechanism of action?

- ☐ 3% diclofenac sodium gel
- ☐ 5% 5-fluorouracil cream
- ☐ Ingenol mebutate
- ☐ Topical imiquimod cream
- ☐ Topical 0.05% tretinoin cream

- ☐ 3% diclofenac sodium gel
- ☒ 5% 5-fluorouracil cream ← Correct answer
- ☐ Ingenol mebutate
- ☐ Topical imiquimod cream
- ☐ Topical 0.05% tretinoin cream

Explanation:

The aim of field cancerization treatment is to reduce the risk of developing cutaneous squamous cell carcinoma (cSCC). Field cancerization is indicated when there is the presence of two or more actinic keratoses (AKs) on a photodamaged area of the skin. All AKs should be treated since AKs are preinvasive of SCCs. Field-directed treatments are effective in the treatment of visible lesions, subclinical lesions and keratinocyte changes in areas surrounding the visible lesions. The treatment options that are approved by the FDA include 5-fluorouracil (5-FU), imiquimod, diclofenac and ingenol mebutate.

5-FU is a pyrimidine analogue that works as a thymidylate synthase inhibitor. The inhibition of this enzyme blocks the synthesis of pyrimidine thymine which is required for DNA replication. Imiquimod acts as a toll-like receptor-7 (TLR-7) agonist which stimulates the innate and adaptive immune response resulting in antitumour activity. Diclofenac inhibits the synthesis of prostaglandin by inhibiting cyclooxygenase enzyme which reduces UV-induced tumourigenesis. The mechanism of action of ingenol mebutate includes mitochondrial disruption leading to necrosis and localised inflammatory response. Tretinoin is not approved by the FDA and its use on field cancerization is still controversial.